

Mathematical pendulum

Homework 3

Marko Zupančič Muc

Contents

1. Mathematical model of the pendulum	1
1.1. Converting to first-order system	1
1.2. DOPRI5	2
2. Numerical experiments	2
2.1. Determining the period	4

1. Mathematical model of the pendulum

The angular motion of a mathematical pendulum is described by

$$\theta''(t) + \frac{g}{l} \sin(\theta(t)) = 0, \quad \theta(0) = \theta_0, \quad \theta'(0) = \omega_0, \quad (1)$$

where l is the pendulum length, g is the gravitational acceleration, θ_0 is the initial angle, and ω_0 is the initial angular velocity.

We solve this equation numerically with the DOPRI5 method which is an adaptive Runge-Kutta method. Because Runge-Kutta methods are designed for first-order differential equations, we must first rewrite the second-order equation as a first-order system.

1.1. Converting to first-order system

Let

$$\omega(t) = \theta'(t) \quad (2)$$

denote the angular velocity.

It follows that

$$\omega'(t) = \theta''(t) = -\frac{g}{l} \sin(\theta(t)). \quad (3)$$

The resulting first-order system is

$$\begin{aligned} \theta'(t) &= \omega(t), \\ \omega'(t) &= -\frac{g}{l} \sin(\theta(t)). \end{aligned} \quad (4)$$

We combine the two variables into the state vector

$$\mathbf{u}(t) = \begin{pmatrix} \theta(t) \\ \omega(t) \end{pmatrix}. \quad (5)$$

Runge-Kutta methods solve systems of the form $\mathbf{u}'(t) = \mathbf{f}(t, \mathbf{u}, \mathbf{p})$. For the pendulum,

$$\mathbf{u}'(t) = \mathbf{f}(t, \mathbf{u}, \mathbf{p}) = \begin{pmatrix} \omega \\ -\frac{g}{l} \sin(\theta) \end{pmatrix}, \quad \mathbf{u}(0) = \begin{pmatrix} \theta_0 \\ \omega_0 \end{pmatrix}, \quad (6)$$

where the parameters are grouped into $\mathbf{p} = (g, l)$.

1.2. DOPRI5

Unlike Euler's method, a Runge-Kutta method evaluates several intermediate slopes during each step. The i -th slope is calculated as

$$k_i = f\left(t_n + c_i h, \mathbf{y}_n + h \sum_{j < i} a_{ij} k_j\right), \quad (7)$$

and the slopes are combined into a weighted sum:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h \sum_i b_i k_i, \quad (8)$$

where b_i are the weights.

The coefficients c_i , a_{ij} , and b_i are commonly arranged in a Butcher tableau.

DOPRI5 is an embedded seven-stage Runge-Kutta method that calculates fifth and fourth-order approximations at the same time. Their difference estimates the local numerical error:

$$\mathbf{e}_{n+1} = \mathbf{y}_{n+1}^5 - \mathbf{y}_{n+1}^4. \quad (9)$$

If the estimated error is too large, the step is rejected and repeated with a smaller value of h . Otherwise, the step is accepted and the method selects an appropriate size for the next step.

The DOPRI5 coefficients are listed in the Butcher tableau in the [Dormand-Prince method reference](#).

2. Numerical experiments

The experiments use $g = 9.81 \text{ m/s}^2$, $l = 1 \text{ m}$, $m = 1 \text{ kg}$, and initial angular velocity $\omega_0 = 0$. The harmonic period associated with these parameters is

$$T_0 = 2\pi \sqrt{\frac{l}{g}} \approx 2.00607 \text{ s}. \quad (10)$$

First, we compare the mathematical pendulum computed with DOPRI5 with the harmonic approximation. The harmonic model replaces $\sin(\theta)$ by θ :

$$\theta_{h''} + \frac{g}{l} \theta_h = 0. \quad (11)$$

For $\omega_0 = 0$, the harmonic solution is¹

$$\theta_{h(t)} = \theta_0 \cos\left(\sqrt{\frac{g}{l}} t\right), \quad (12)$$

with the period T_0 given above.

We solve the nonlinear model for three initial angles:

$$\theta_0 = 0.1, 1.0, 2.5. \quad (13)$$

Each solution is computed over three harmonic periods.

DOPRI5 chooses its time steps adaptively, so different solutions do not generally contain values at the same times. For plotting, we evaluate them on one common uniform grid. Intermediate values are obtained with cubic Hermite interpolation, following the approach used in Vajé 16². The comparison is shown in Figure 1.

¹[https://en.wikipedia.org/wiki/Pendulum_\(mechanics\)#Small-angle_approximation](https://en.wikipedia.org/wiki/Pendulum_(mechanics)#Small-angle_approximation)

²https://gitlab.com/nummat/nummat-knjiga/-/blob/master/Vaja16/src/Vaja16.jl?ref_type=heads

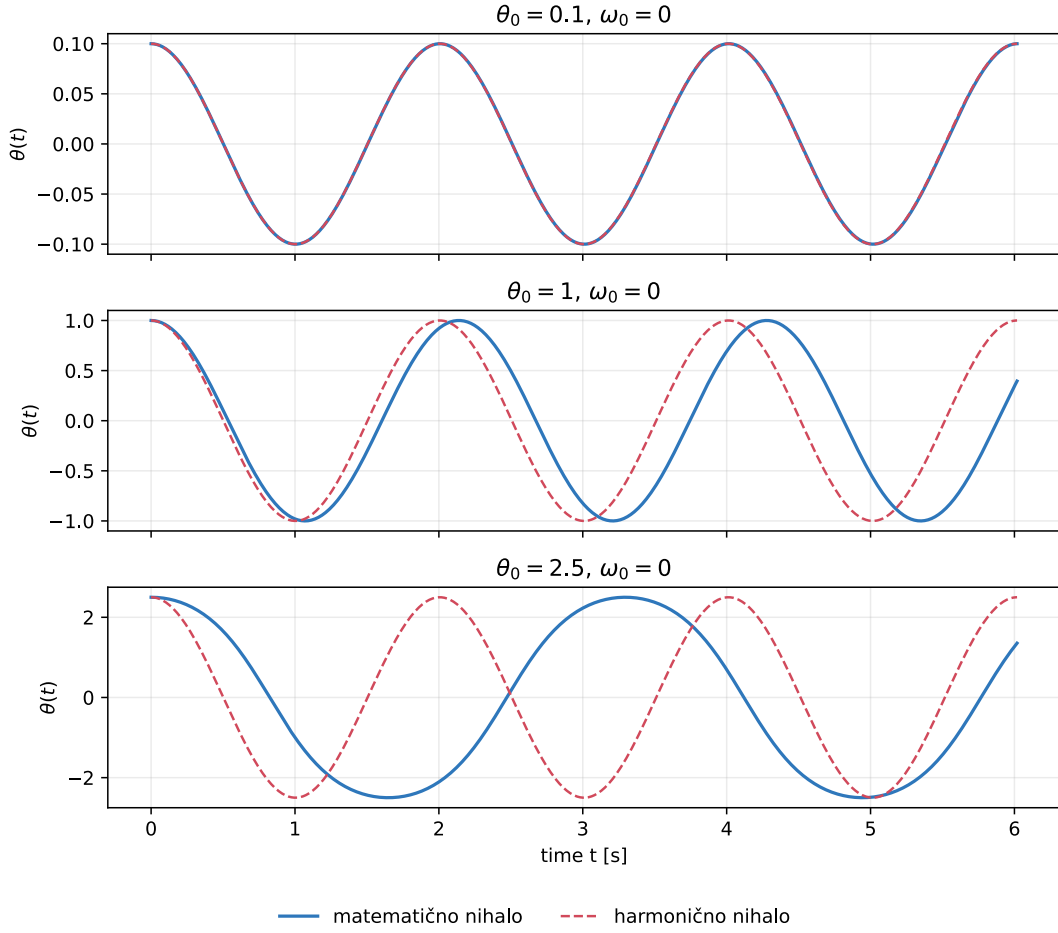


Figure 1: Comparison of the nonlinear mathematical pendulum computed with DOPRI5 and the harmonic approximation for three initial angles.

For the smallest angle $\theta_0 = 0.1$, the two curves are almost identical. This is expected because $\sin(\theta) \approx \theta$ near zero.

At $\theta_0 = 1.0$, the difference between the periods becomes visible and gradually produces a shift.

At $\theta_0 = 2.5$, the harmonic approximation is no longer accurate, as it oscillates considerably faster than the nonlinear pendulum. For $0 < \theta < \pi$, we have $\sin(\theta) < \theta$, so the restoring acceleration in the nonlinear model is weaker than in the harmonic model. The nonlinear pendulum consequently needs more time to complete one oscillation.

The second experiment examines the relationship between the period of the nonlinear pendulum and its energy. The graph is not constructed from the time steps of a single trajectory. Instead, we repeat the complete simulation for many initial angles. Each initial angle produces one energy value and one measured period, giving one point (E, T) in the graph.

The mechanical energy is³

$$E(t) = \frac{1}{2}ml^2\omega(t)^2 + mgl(1 - \cos(\theta(t))). \quad (14)$$

Because every pendulum is released from rest, $\omega_0 = 0$, its initial energy simplifies to

$$E_0 = mgl(1 - \cos(\theta_0)). \quad (15)$$

Since the model contains no damping, the total energy should remain constant during one simulation. Consequently, we calculate one energy value for each initial condition rather than treating the DOPRI5 time points as different energy levels. The graph uses the dimensionless quantity $\frac{E}{mgl}$.

³[https://en.wikipedia.org/wiki/Pendulum_\(mechanics\)](https://en.wikipedia.org/wiki/Pendulum_(mechanics)) and https://en.wikipedia.org/wiki/Spherical_pendulum

2.1. Determining the period

For every initial angle, DOPRI5 first produces a numerical trajectory $(\theta(t), \omega(t))$. We determine its period by detecting zero crossings of $\theta(t)$. Starting from a positive turning point with $\omega_0 = 0$, the pendulum first crosses $\theta = 0$ while moving downwards. After one complete oscillation it crosses $\theta = 0$ downwards again. The difference between the times of these two crossings is therefore the period:

$$T = t_2 - t_1. \quad (16)$$

The zero search is performed in two stages. First, consecutive stored DOPRI5 states are inspected until the angle changes sign, which means we've bracketed a zero. The bracket is then narrowed with bisection. At each trial time, the state is reconstructed from the left endpoint with a DOPRI5 step.

The procedure is repeated for initial angles between 0.05 and 0.95π . For every angle we store

$$(E_0, T). \quad (17)$$

The resulting graph is shown in Figure 2.

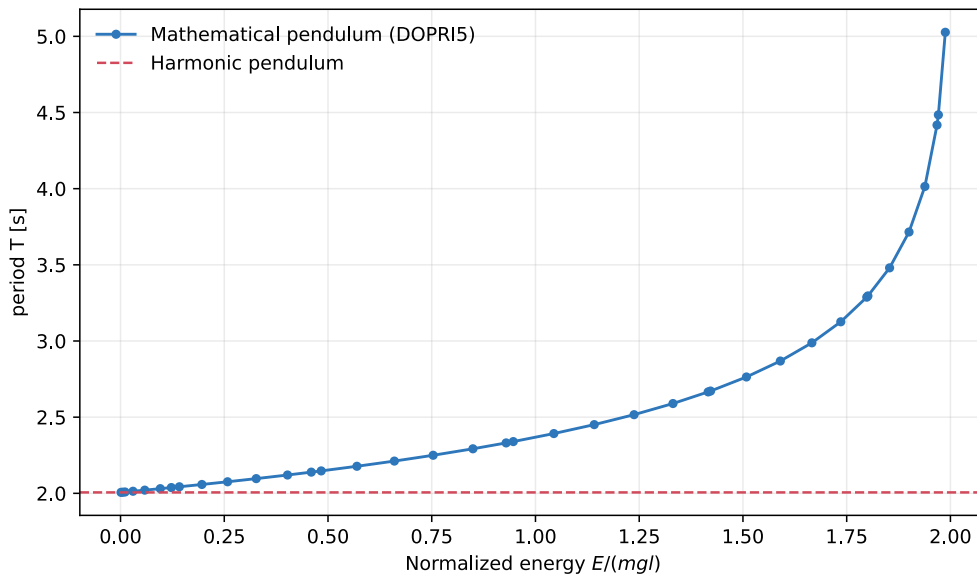


Figure 2: Period of the mathematical pendulum as a function of normalized initial energy. The dashed line is the constant harmonic period T_0 .

At low energy, the mathematical pendulum is close to the harmonic model, so its period is close to T_0 . As the initial energy and thus amplitude increase, the nonlinear period increases. The harmonic line stays constant because the harmonic period is independent of amplitude and energy. Near $\frac{E}{mgl} = 2$, the initial angle approaches the unstable upright position $\theta_0 = \pi$. The pendulum then spends increasingly more time near the turning point, which explains the steep rise of the curve.